Banking & Financial Services

BM

Bank Marketing *Campaign* Analytics

Optimising term-deposit conversion through customer, campaign & macro-economic analytics — across 41,176 direct-marketing contacts.

41,176	11.27%	6.9×
CONTACTS ANALYSED	BASELINE CONVERSION	TOP-DRIVER LIFT

PROJECT
Bank Marketing Campaign Analytics
UCI Portuguese Bank Dataset

SECTOR
Banking & Financial Services
Retail · Direct Marketing

GROUP
Group — 11

SUBMISSION
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CONTENTS

Table of Contents

01	Executive Summary	04
02	Sector & Business Context	06
03	Problem Statement & Objectives	07
04	Data Description	09
05	Cleaning & Transformation	11
06	KPI Framework	13
07	Exploratory Analysis	16
08	Statistical Analysis	19
09	Dashboard Walkthrough	22
10	Key Insights	28
11	Business Recommendations	30
12	Limitations & Next Steps	32
13	Contribution Matrix	33

Part I

Foundation

Setting the strategic ground — the sector, the business problem, and the analytical objectives that define the rest of this work.

01 Executive Summary
02 Sector & Business Context
03 Problem Statement

I

lightweight propensity model on the six features identified by the logistic-regression step (90.8% accuracy on the cleaned dataset).

SECTION 02

Sector & Business Context

Sector Overview

Retail banking in Europe relies heavily on **term deposits** as a stable funding source for the loan book. Term-deposit campaigns are typically run by an outbound call centre and target the bank's existing customer base. Two characteristics define the economics of these campaigns: (i) labour cost per contact is high relative to the marginal interest spread on a single deposit, and (ii) conversion rates are typically in single digits, so even small percentage-point gains in conversion materially change campaign ROI. The UCI Bank Marketing dataset captures one such campaign by a Portuguese banking institution and is widely used as a benchmark for marketing analytics work.

> *At an 11.27% baseline conversion rate, roughly 89 of every 100 calls produce no revenue but consume agent time. Even modest segment-level gains compound into significant ROI shifts.*

Decision-Maker / Stakeholder

The primary stakeholder is the **Head of Direct Marketing / Campaign Manager** at the bank, supported by the analytics function and the call-centre operations team. Secondary stakeholders include the Treasury team (who fund the deposit book), the Compliance team (who govern outbound-contact frequency under consumer-protection rules), and Branch Managers in high-conversion regions.

Why This Problem Matters

If the analysis can lift conversion in targeted segments to even 20% while reducing wasted contacts, the bank gains on three dimensions: lower cost per acquired deposit, faster campaign cycles, and reduced consumer fatigue from repeated calls. The macro-economic features in the dataset further allow timing decisions — running heavy campaigns in low-interest-rate windows when customer responsiveness peaks at **24.02%**.

SECTION 03

Problem Statement & Objectives

Formal Problem Definition

Given historical contact-level records of a Portuguese bank's term-deposit marketing campaign — comprising customer demographics, contact metadata, prior-campaign history and macro-economic indicators — **identify** the customer, campaign and economic factors that are statistically associated with subscription outcome, and **quantify** their relative impact so that the campaign manager can re-prioritise outreach.

Scope

- **In scope:** descriptive analytics, exploratory analysis, hypothesis testing and explanatory logistic regression on the cleaned dataset; segment-level subscription rates; Tableau dashboards across four analytical themes (Customer, Campaign, Interaction History, Economic).
- **Out of scope:** production deployment of a predictive model, real-time scoring, A/B test design, customer-lifetime-value modelling, and any work requiring data outside the UCI dataset (e.g., transaction logs, channel costs, call-recording transcripts).

Success Criteria

- Reproducible cleaning pipeline that yields a single canonical analysis dataset.
- All major hypotheses about subscription drivers tested at *alpha* = 0.05 with appropriate methods.
- **Four interactive dashboards** covering customer, campaign, interaction-history and economic perspectives.
- A logistic-regression baseline with explainable coefficients and accuracy ≥ 85%.
- Concrete, ranked recommendations linked back to specific statistical evidence.

Part II

Data & Measurement

How the dataset was prepared, what was engineered, and the KPI framework that governs every downstream view.

04 Data Description
05 Cleaning & Transformation
06 KPI Framework

II

SECTION 04

Data Description

Source & Access

The data is the **Bank Marketing Data Set** originally collected by Moro, Cortez & Rita (2014) and hosted on the **UCI Machine Learning Repository**. It captures direct-marketing phone campaigns from a Portuguese banking institution, with the binary target indicating whether the contacted customer subscribed to a term deposit.

Access link: archive.ics.uci.edu/dataset/222/bank+marketing

Dataset Size and Coverage

- **Raw shape:** 41,188 rows × 21 columns (one row per customer-contact event).
- **Cleaned shape:** 41,176 rows × 27 columns (12 duplicate rows dropped; 7 engineered features added; 1 raw recency column dropped; *region* NUTS II column added).
- **Target:** *subscription_flag* — 1 = subscribed, 0 = did not subscribe. Overall conversion rate = **11.27%** (highly imbalanced).
- **Time / macro context:** the macro-economic columns (employment-variation rate, CPI, CCI, Euribor 3-month rate, number of employees) span a multi-year window covering the 2008 financial-crisis aftermath, providing meaningful variance in interest-rate environment.

Key Columns

Column	Type	Description
age	Numeric	Customer age in years (17–98).
job_type	Categorical	12 occupation categories (admin, blue-collar, technician, etc.).
marital_status / education_level	Categorical	Demographic context.
contact_type	Categorical	cellular vs telephone.
contact_month / contact_day	Categorical	Month and weekday of contact.
call_duration_sec	Numeric	Length of the contact call in seconds.
num_contacts_campaign	Numeric	Calls within current campaign (capped at 10).
num_previous_contacts	Numeric	Contacts before this campaign.

Column	Type	Description
previous_campaign_outcome	Categorical	success / failure / nonexistent.
Macro indicators *employment_variation_rate, CPI, CCI, interest_rate, number_of_employees*	Numeric	Quarterly economic indicators capturing the rate environment at contact time.
subscription_flag	Binary (target)	1 if customer subscribed, else 0.

Data Quality Issues Identified

- **No null values** in any column (verified via *df.isnull().sum()*).
- **12 fully duplicate rows** — removed in cleaning.
- **Outliers in num_contacts_campaign:** a small number of records had extreme call counts (up to 56). Capped at 10 to prevent distortion of aggregations.
- **Misleading placeholder in days_since_last_contact:** the value 999 indicated 'never contacted' but skewed any numeric summary. Replaced with a categorical *contact_recency* field.
- **Tiny category 'illiterate'** in education had too few records for stable inference — merged into 'basic.4y'.
- **No native geography**: the raw dataset has no region. We added a NUTS II *region* column via probability-based imputation conditioned on job_type.

BUSINESS LOGIC NOTE — 'UNKNOWN' VALUES

Several categorical fields (job_type, marital_status, education_level, credit_default, has_housing_loan, has_personal_loan) contain the value 'Unknown'. **This is not a data error** — it reflects customers for whom the bank genuinely does not hold that piece of information. These records were retained as-is and the 'Unknown' category was treated as a valid segment in its own right. In several analysis views, 'Unknown' education and 'Unknown' credit default show distinct conversion patterns, confirming that the absence of information is itself a meaningful signal.

SECTION 05

Cleaning & Transformation

Major Cleaning Steps

1 **Load & inspect**
Loaded semicolon-separated raw file; ran *info()*, *describe()*, *nunique()* as a baseline check.

2 **De-duplicate**
Dropped 12 duplicate rows → 41,176 unique records.

3 **Type fix**
Cast *nr_employed* to integer for consistent arithmetic.

4 **Rename columns**
Standardised cryptic names (e.g., *nr_employed* → *number_of_employees*).

5 **Standardise values**
Target yes/no → 1/0; literal 'unknown' → 'Unknown' across categoricals.

6 **Merge tiny category**
illiterate rolled into *basic.4y* (too few records).

7 **Categorical typing**
Explicit *category* dtype on relevant columns (memory + Tableau).

8 **Cap outliers**
num_contacts_campaign clipped at 10.

9 **Engineer age_group**
5 buckets: Young ≤25 / Young-Mid 26–35 / Mid 36–45 / Mature 46–55 / Senior 55+.

10 **Engineer call_intensity**
Buckets: 1 / 2 / 3–5 / 6+ contacts.

11 **Engineer contacted_before**
Yes / No based on *num_previous_contacts*.

12 **Engineer contact_recency**
999 → 'Never Contacted'; ≤7 / ≤30 / >30 buckets.

13 **Drop raw recency**
Removed *days_since_last_contact* after replacement.

14 **Engineer rate_environment**
Low / Medium / High based on *interest_rate*.

15 **Engineer prev_success_flag**
Binary flag from *previous_campaign_outcome*.

16 **Engineer education_ordinal**
Numeric mapping respecting education hierarchy.

17 **Add region (NUTS II)**
Probability-based imputation conditioned on *job_type* to enable regional analysis.

Assumptions Made

- The 'Unknown' category across demographic fields reflects genuine gaps in the bank's customer records — not data entry errors. These customers are retained as a distinct segment because their conversion behaviour differs measurably from fully-profiled customers.
- Capping campaign calls at 10 — chosen because higher counts are rare and operationally unusual; the underlying signal (overcalling hurts conversion) is preserved.

- Education ordinality assumes a strict basic → high.school → professional.course → university.degree hierarchy.
- **Region imputation:** assigned each record a NUTS II region by sampling from a *P(region | job_type)* distribution. This adds geographic stratification for dashboarding but is a synthetic enrichment, not ground truth.

Output Dataset

The cleaning pipeline produces **Bank_Marketing_Dataset_Cleaned_Data.csv** — 41,176 rows × 27 columns — used downstream by EDA, statistical analysis and the four Tableau dashboards. All engineered fields are designed to be self-explanatory and Tableau-friendly.

SECTION 06

KPI Framework

Each dashboard surfaces a small set of headline KPIs. The framework below codifies their definition, formula and business rationale. Values shown are computed on the full cleaned dataset (n = 41,176).

CONVERSION RATE

11.27%

subscribed / contacts × 100

The single headline KPI. Every segment is benchmarked against it.

SUCCESSFUL CUSTOMERS

4,639

Count where subscription_flag = 1

Absolute acquisitions — feeds deposit-book growth.

AVG CALL DURATION

4.3 sec

Mean(call_duration_sec)

Behavioural signal of engagement; strongest correlate (r = +0.41).

AVG CONTACTS / CUSTOMER

2.44

Mean(num_contacts_campaign)

Effort per customer. Crossing 3 calls reduces yield.

CELLULAR CONVERSION

14.74%

subscribed / contacts (cellular)

Channel-level KPI; cellular outperforms landline ~3×.

PRIOR-SUCCESS CONVERSION

65.11%

subscribed / contacts (prev=success)

Loyalty KPI — the dominant predictor (6.9× lift).

COLD-CONTACT CONVERSION

8.83%

subscribed / never-contacted

Floor benchmark for unfamiliar prospects.

LOW-RATE ENV. CONV.

24.02%

subscribed / contacts (rate=Low)

Macro-timing KPI; ~5× the high-rate yield.

HIGH-RATE ENV. CONV.

4.83%

subscribed / contacts (rate=High)

Cost of mistimed campaigns — worst-case benchmark.

AVG INTEREST RATE

3.62%

Mean(interest_rate)

Macro context for the campaign window.

AVG CONSUMER CONF. IDX

−40.50

Mean(consumer_confidence_index)

Sentiment proxy; correlates positively (r = +0.05).

KPI-to-Dashboard Mapping

Dashboard	Headline KPIs
Customer Profile	Total Customers, Customer Contacted, Successful, Unsuccessful, Success Rate
Campaign Effectiveness	Conversion Rate, Avg Call Duration, Avg Contacts Per Customer, Cellular Conversion
Interaction History	Previously Contacted, Prior Success, Prior-Success Conversion, Cold-Contact Conversion
Economic Impact	Low-Rate Conv., High-Rate Conv., Avg Interest Rate, Avg Consumer Confidence Index

Part III

Analysis

Patterns in the data, validated by statistics and surfaced through interactive dashboards.

07 Exploratory Analysis
08 Statistical Analysis
09 Dashboard Walkthrough

III

SECTION 07

Exploratory Analysis

Major Trends

- **Class imbalance dominates.** Only 11.27% of contacts subscribe — every analysis must account for the rare-event nature of the target.
- **Age is concentrated 30–50** with a long right tail; call duration is heavily right-skewed; most customers are called only 1–3 times within a campaign.
- **Volume vs yield mismatch.** May has the highest contact volume but March, September and December produce the highest conversion rates — the campaign is busiest in a low-yield month.
- **Cellular >> telephone.** Cellular contacts convert at 14.74% versus 5.23% for landline.

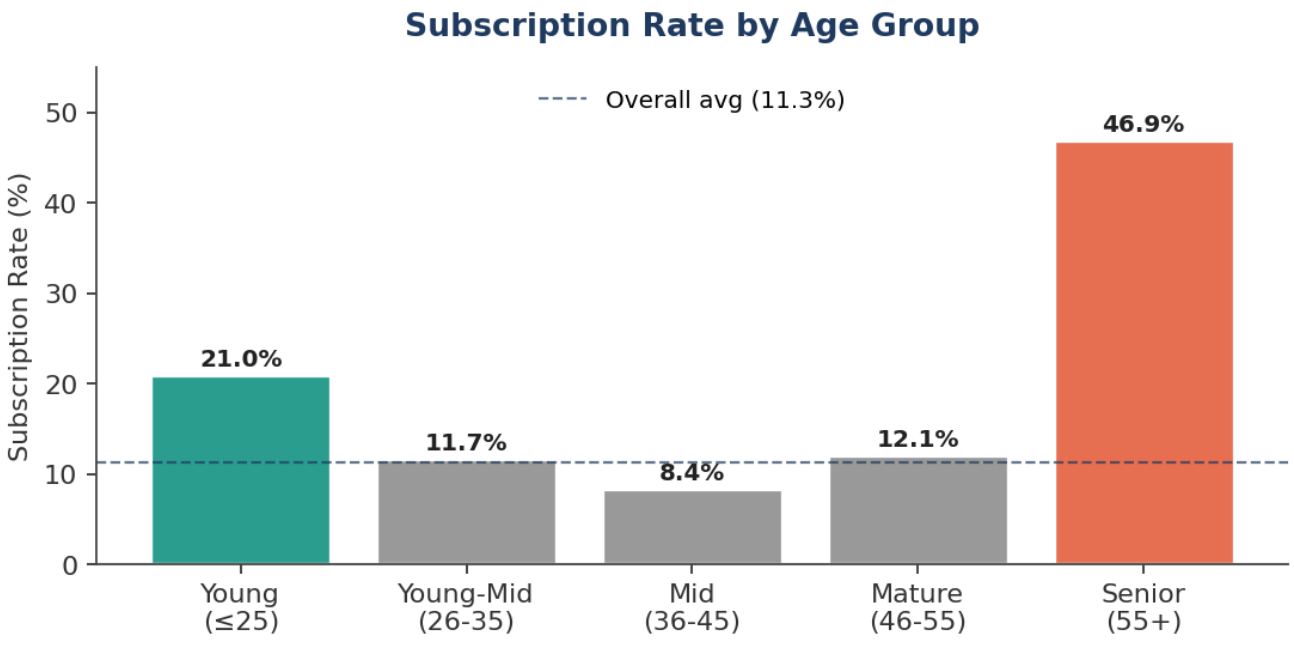


FIGURE 1 *Subscription rate by age group. The relationship is U-shaped: Senior (55+) and Young (≤25) cohorts substantially outperform the 36–45 middle, which sits below the 11.27% overall average.*

Job-type analysis reinforces the demographic signal: **students** and **retired** customers convert at materially higher rates than blue-collar workers. Higher education (university degree) gives a smaller but consistent lift over basic education levels.

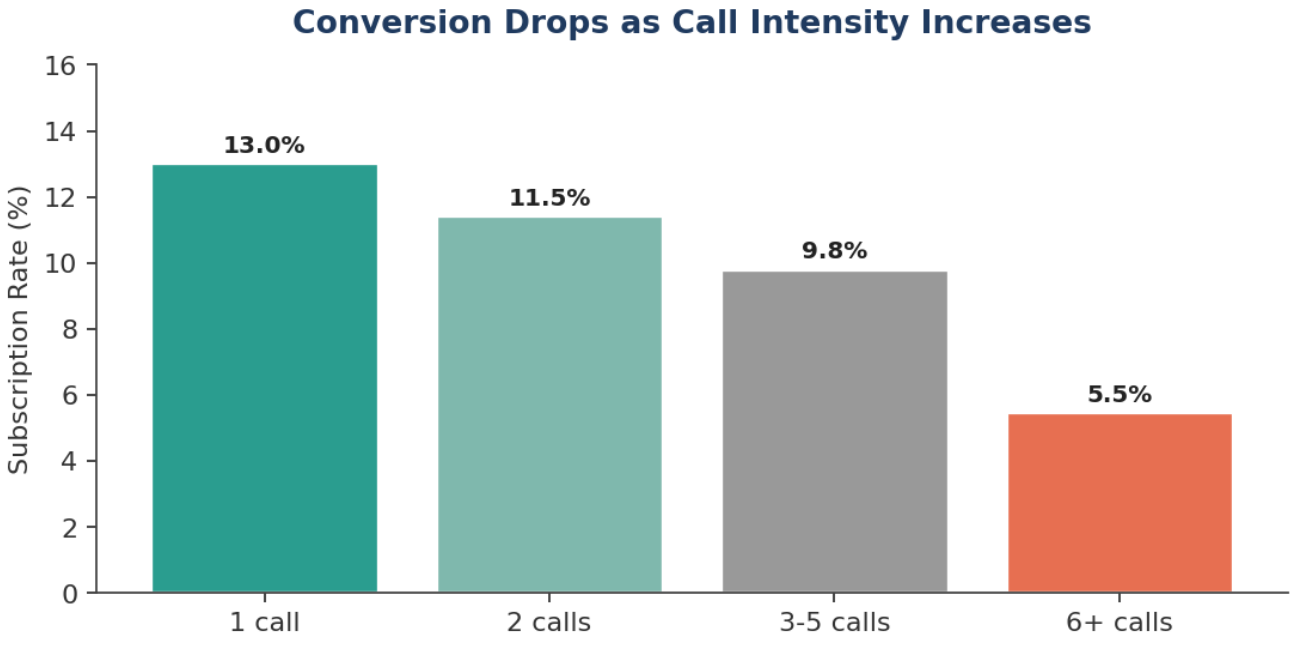


FIGURE 2 *Subscription rate by call intensity. Conversion declines monotonically: each additional call reduces yield, with 6+ contacts producing less than half the conversion of a single call.*

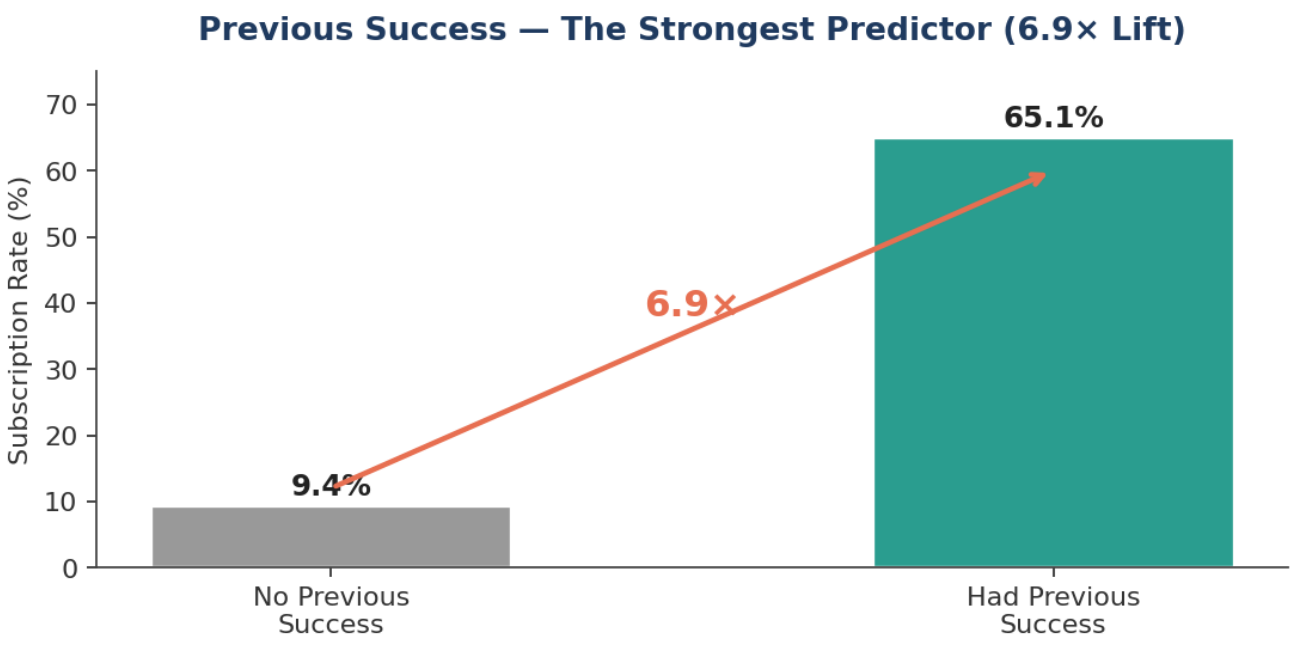


FIGURE 3 *Customers who subscribed in a prior campaign are 6.9× more likely to subscribe again — by far the largest segment-level gap observed in the dataset.*

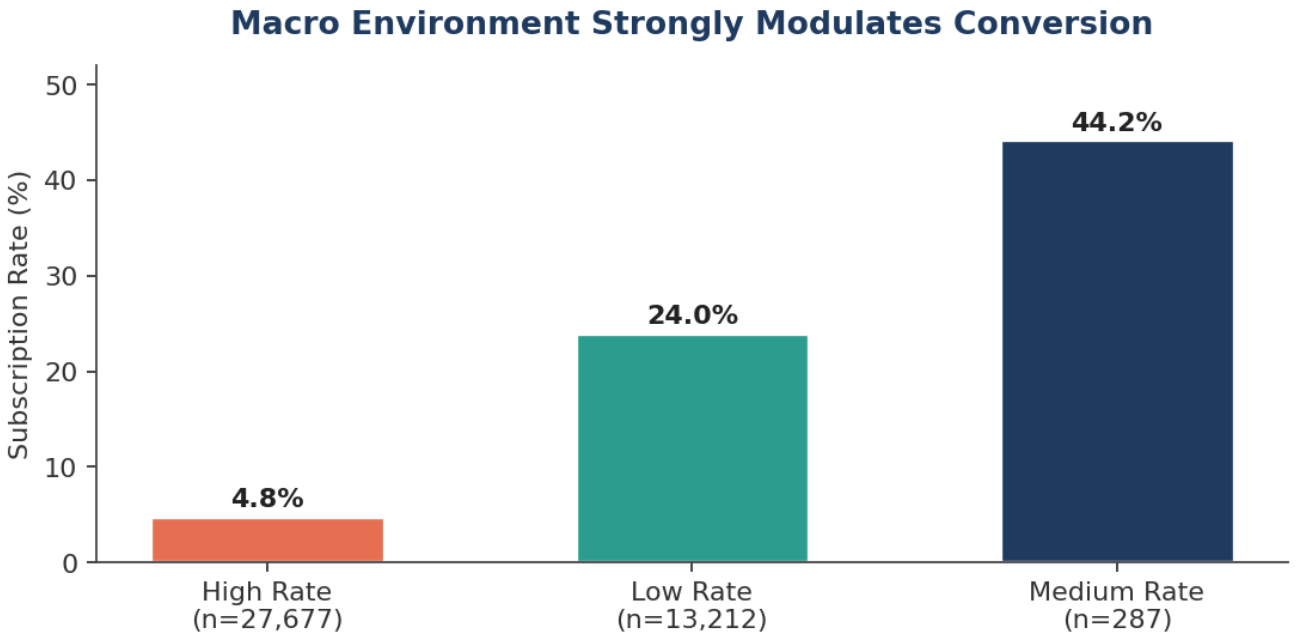


FIGURE 4 *Conversion in low- and medium-rate environments runs 5–9× higher than in high-rate windows. Macro timing is one of the largest controllable levers in this dataset.*

Outlier & Distribution Notes

- **num_contacts_campaign:** capped at 10 in cleaning — extreme values would otherwise distort segment averages.
- **call_duration_sec:** many statistical outliers exist (very long calls), but these are genuine and highly predictive — retained intentionally.
- **Macro indicators** are highly mutually correlated (employment-variation rate, CPI, interest rate, employee count) — they jointly describe one underlying economic regime.

SECTION 08

Statistical Analysis

Methods Used

We tested six hypotheses linked to the four dashboard themes. ANOVA was used to compare subscription rates across 3+ groups (age, call intensity, rate environment); Chi-Square Tests of Independence were used for two categorical variables (job, contact type, prior success). All tests were run at *alpha* = 0.05. Effect sizes were then quantified with Pearson correlation and a multivariate logistic-regression model (six features, 90.8% accuracy).

Hypothesis-Testing Results

Factor	Test	Statistic	P-Value	Result & Dashboard
Age Group	ANOVA	**281.01**	< 0.001	SIG. Customer Profile
Job Type	Chi-Square	**961.74**	< 0.001	SIG. Customer Profile
Call Intensity	ANOVA	**63.01**	< 0.001	SIG. Campaign
Contact Type	Chi-Square	**862.08**	< 0.001	SIG. Campaign
Previous Success	Chi-Square	**4,113.95**	< 0.001	HIGHLY SIG. Interaction History
Rate Environment	ANOVA	**1,977.46**	< 0.001	SIG. Economic

"Highly Sig." flags an extremely strong result — Chi-Square statistic an order of magnitude larger than other tests.

> *All six tests reject the null hypothesis at $p < 0.001$. The Chi-Square statistic for previous-campaign success (≈ 4,114) is an order of magnitude larger than any other test — confirming prior success as the dominant signal.*

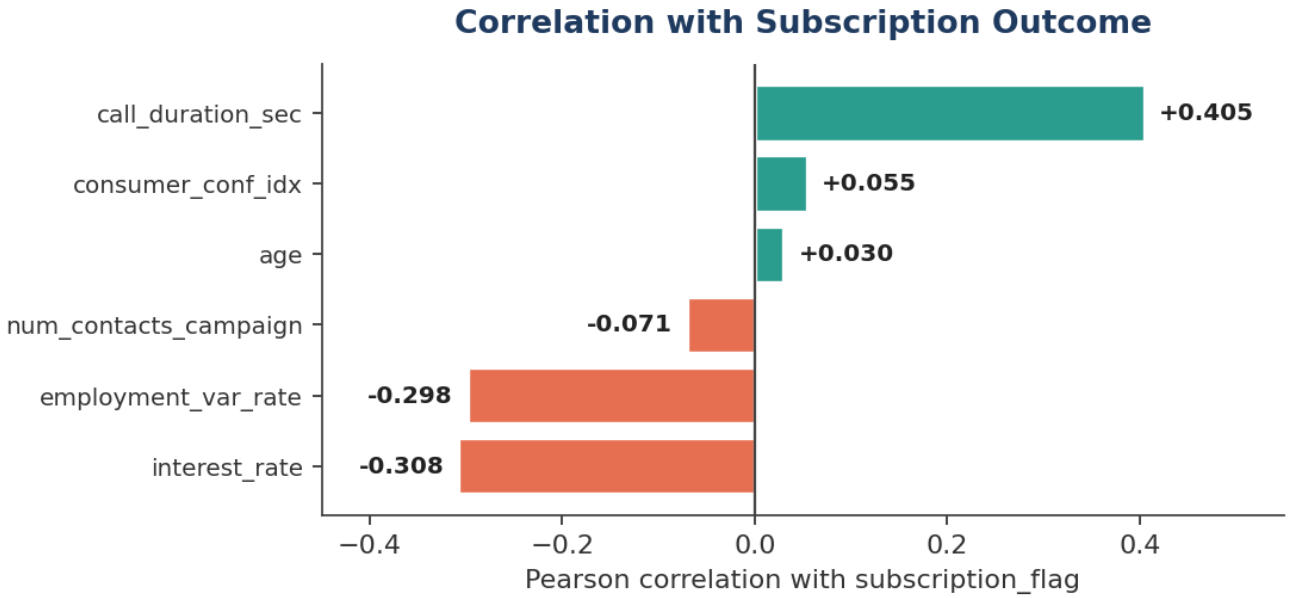


FIGURE 5 *Pearson correlation of numeric features with subscription_flag. call_duration_sec is the strongest positive driver (+0.41); interest_rate and employment-variation rate are the strongest negatives.*

Logistic Regression — Feature Impact

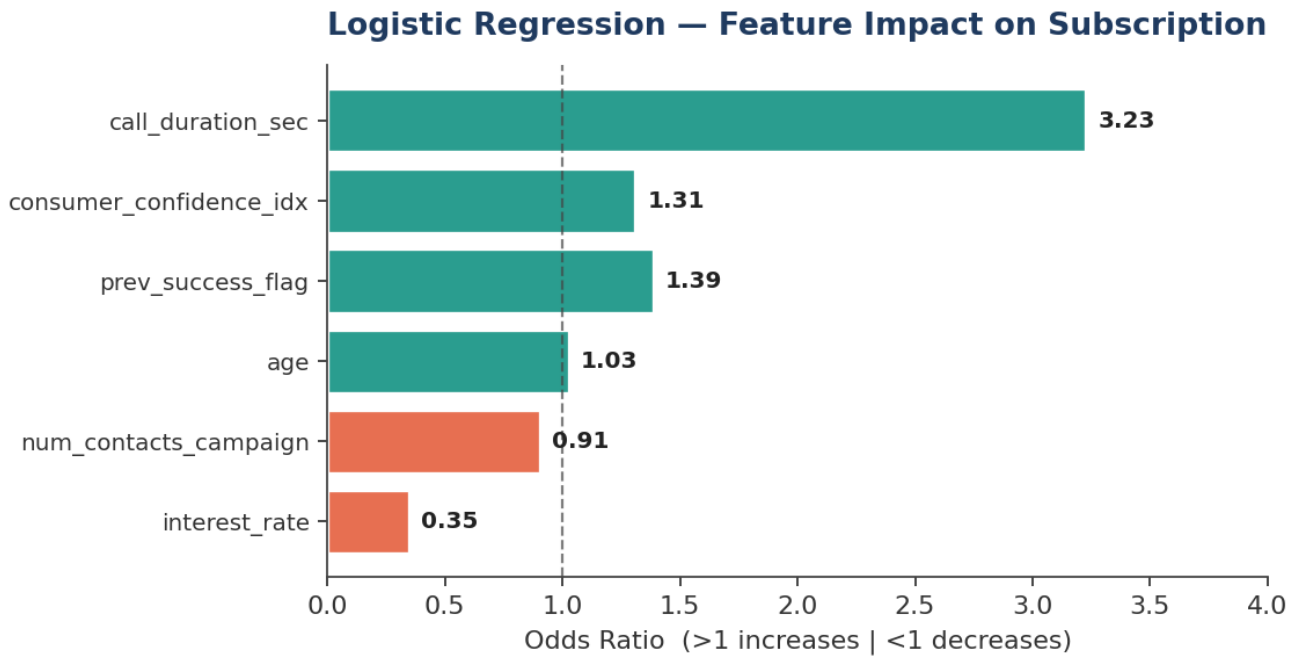


FIGURE 6 *Odds ratios from the multivariate logistic regression. Call duration triples the odds of subscribing (OR = 3.23), while a high-rate environment cuts them by ~65% (OR = 0.35). Model accuracy:* **90.8%**.

Business Interpretation

- **Behavioural signals matter most.** Call duration is the strongest in-conversation signal of intent — agents should be trained to extend qualified conversations rather than push for fast resolution.
- **Macro timing is exogenous but exploitable.** Interest-rate environment is the second-largest driver and is observable in advance — campaign volume should ramp in low-rate windows.
- **Past success is a predictive gold mine.** A prior-success flag should become a first-class field in any future campaign list — it carries a 6.9× lift on its own.
- **Effort is not the answer.** Both correlation and the logistic-regression coefficient on *num_contacts_campaign* are negative — calling more does not help, it actively hurts.

SECTION 09

Dashboard Walkthrough

Dashboard Objective

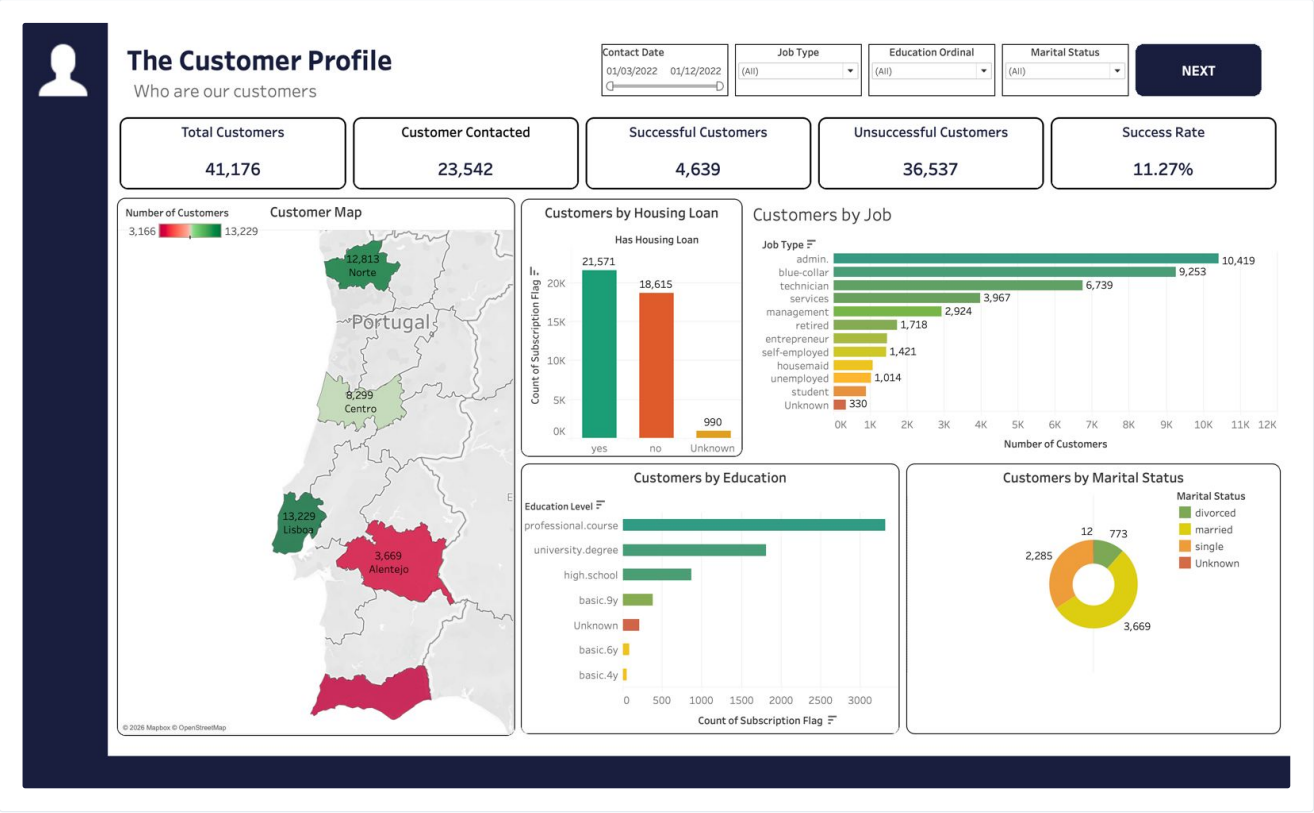
Four interactive Tableau dashboards translate the analytical findings into decision-ready views for the campaign manager. The dashboards are organised around the same four themes used in the statistical analysis — Customer Profile, Campaign Effectiveness, Interaction History and Economic Environment — so each visual ties directly back to a hypothesis test in Section 08.

Filters & Interactivity

- **Cross-dashboard filters:** Contact Date, Job Type, Education Ordinal, Marital Status — consistent across all four dashboards.
- **Navigation:** Previous / Next buttons step through the four dashboards as a guided story.
- **KPI tiles:** top-of-page tiles surface the headline metric for each dashboard alongside three to four supporting KPIs.
- **Tooltips:** hover on any bar to surface sample size — important given the rare-event target.

Dashboard 1 — Customer Profile

Who are our customers? Demographic distribution by region, job, education and marital status across all 41,176 contacts.



41,176	23,542	4,639	36,537	11.27%
TOTAL CUSTOMERS	CONTACTED	SUCCESSFUL	UNSUCCESSFUL	SUCCESS RATE

KEY OBSERVATIONS FROM DASHBOARD 1

- **Norte (12,813) and Lisboa (13,229)** together account for over 63% of all customers, making them the primary target regions for any concentrated campaign push.
- **Alentejo (3,669)** is the smallest region by volume — any conversion gains here will have limited aggregate impact, but regional benchmarking may still reveal high per-contact efficiency.
- **Admin (10,419) and blue-collar (9,253)** are the two largest job segments; however, retired and student customers — despite being smaller — carry significantly higher conversion rates, creating a clear targeting opportunity.
- **Housing loan holders (21,571)** outnumber non-holders (18,615) — this existing financial relationship with the bank may correlate with greater product receptivity, warranting investigation of cross-sell potential.
- **Professional.course and university.degree** dominate the education distribution, suggesting the average customer is relatively well-educated — messaging should reflect product sophistication rather than over-simplifying term-deposit benefits.
- **Married customers (3,669)** form the largest marital-status segment, followed by single (2,285). Household financial stability associated with married status may explain higher willingness to lock funds into a deposit.

- **11.27% overall success rate** — despite 41,176 total customers, only 4,639 subscribed. This confirms that the default contact strategy is scattershot and strongly supports the case for propensity-based prioritisation.

Dashboard 2 — Campaign Effectiveness

How well are we calling? Subscription rate by call intensity, contact channel, month and prior outcome.

11.27%	4.3	2.44	14.74%
CONVERSION RATE	AVG CALL DURATION	AVG CONTACTS / CUST.	CELLULAR CONVERSION

KEY OBSERVATIONS FROM DASHBOARD 2

- **Call intensity is inversely correlated with conversion:** a single contact converts at 13.04%, dropping monotonically to 5.49% at 6+ calls. Every additional call beyond the third is actively destroying yield — agents are spending time on the least likely-to-convert customers.
- **March (50.55%), December (48.90%) and September (44.91%)** are the highest-converting months by a large margin — more than 4× the conversion of May (6.44%), which is paradoxically the highest-volume contact month.
- **May has the highest contact volume yet the lowest conversion (6.44%)** — this is the clearest operational inefficiency in the dataset. Campaign resources are concentrated in exactly the wrong month.
- **Cellular contacts convert at 14.74% versus telephone at 5.23%** — nearly a 3× advantage. The contact channel is one of the most actionable levers available since it requires no change to customer targeting, only to dialler configuration.
- **Prior-outcome breakdown** shows 35,551 customers were never contacted before (nonexistent), 4,252 had a failed prior campaign, and only 1,373 had a prior success — yet that 1,373 drives a disproportionate share of subscriptions.

- **Recently contacted customers** convert at significantly higher rates (1,177 successes) than never-contacted customers (39,661 in the pool, very low yield rate) — recency of prior engagement is a strong positive predictor of re-engagement.
- **Avg call duration of 4.3 minutes** — longer calls strongly correlate with subscription (r = +0.41). Training agents to sustain qualified conversations rather than terminate quickly could directly lift the conversion rate.

Dashboard 3 — Interaction History

How do interaction patterns show loyalty? Prior campaign history, recency and success flags as predictors of re-subscription.

5,625	1,373	65.11%	8.83%
PREVIOUSLY CONTACTED	PRIOR SUCCESS	PRIOR-SUCCESS CONV.	COLD-CONTACT CONV.

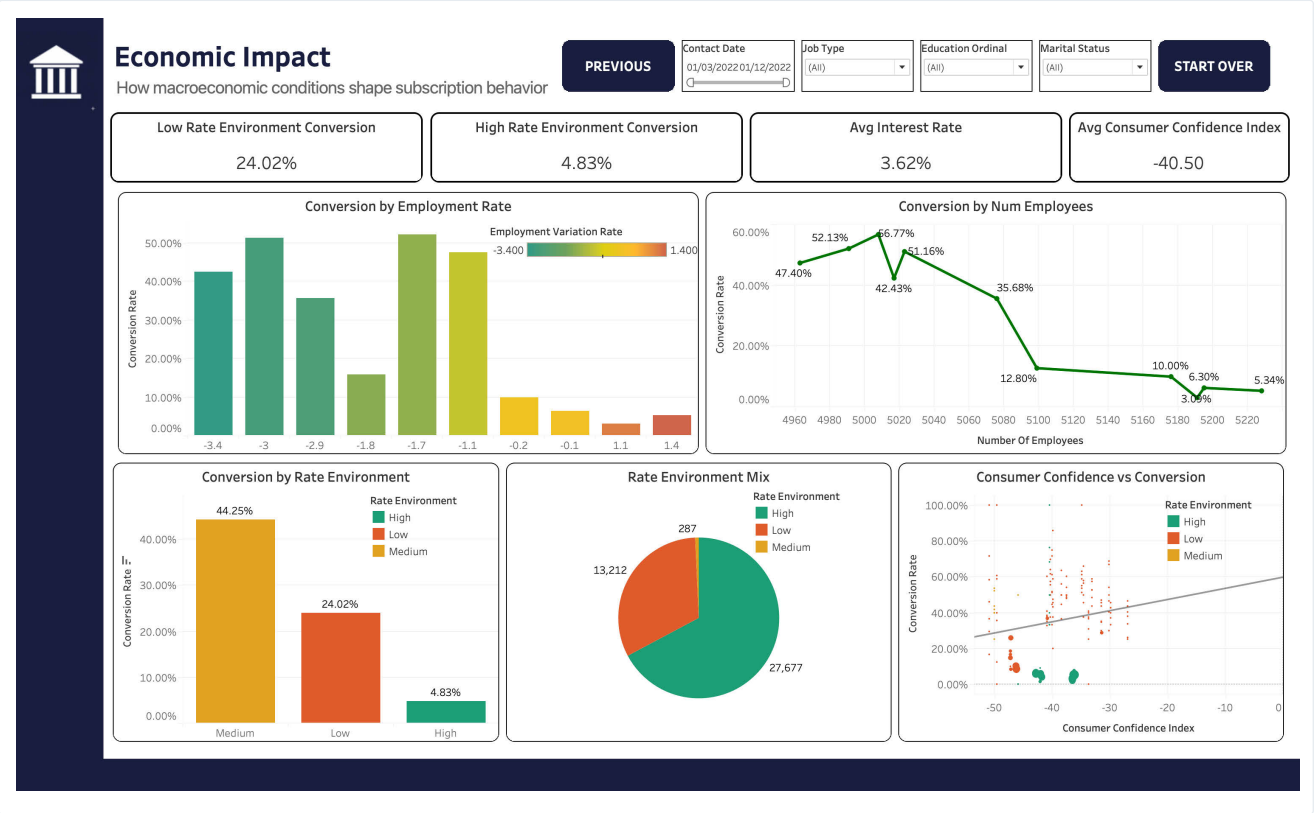
KEY OBSERVATIONS FROM DASHBOARD 3

- **Prior success is the single most powerful predictor:** customers who subscribed in a previous campaign convert again at **65.11%** versus just 8.83% for cold contacts — a 6.9× lift. This is the highest-leverage segment in the entire dataset.
- **Prior failure still beats cold contact:** customers who failed to subscribe previously still convert at 14.23%, 60% higher than the cold baseline. Even a "failed" prior relationship has value worth re-activating.
- **35,551 of all contacts had no prior campaign history** — this is the "never-contacted" majority. Their 8.83% cold conversion rate establishes the floor benchmark and confirms the need for a smarter initial targeting list.
- **Conversion increases with previous contacts up to 5** (peaking at 72.22% at 5 contacts) then drops sharply — a clear sweet spot that suggests sustained prior relationship-building pays off, but there is a natural ceiling.
- **Contact recency drives conversion:** recently contacted customers (1,177 successes) convert at substantially higher rates than moderately-contacted (338) or never-contacted (39,661) groups — freshness of relationship matters as much as its existence.

- **Previous Success Flag** confirms the binary divide: customers with flag = 1 (65.11%) versus flag = 0 (9.41%) — a ~7× gap. This single binary field should be the first-sort criterion on every campaign dialling list.
- **Only 1,373 customers have a prior success flag** out of 41,176 — just 3.3% of the base. This elite segment is under-prioritised relative to its conversion power and represents the lowest-cost route to improving campaign ROI.

Dashboard 4 — Economic Impact

How do macroeconomic conditions shape subscription behaviour? Rate environment, employment trends and consumer sentiment as campaign timing signals.



24.02%	4.83%	3.62%	−40.50
LOW-RATE ENV. CONV.	HIGH-RATE ENV. CONV.	AVG INTEREST RATE	AVG CONSUMER CONF.

KEY OBSERVATIONS FROM DASHBOARD 4

- **Rate environment is the most powerful macro lever:** Medium-rate windows convert at 44.25%, Low at 24.02%, and High at a mere 4.83% — a ~9× difference between the best and worst macro context. Campaign intensity should modulate with the rate environment.
- **High-rate periods dominate contact volume** (27,677 contacts in high-rate vs 13,212 in low-rate) — the bank is doing the most calling when customers are least receptive. This timing inversion is a major structural inefficiency.
- **Employment variation rate strongly predicts conversion:** at EVR = −3.4 (workforce contracting), conversion reaches ~42–52%. As EVR rises toward +1.4 (expanding economy), conversion falls sharply. Customers in an economically uncertain environment are more willing to lock funds into safe deposits.
- **Higher employee counts (5,140+) correlate with lower conversion** (down to 5.34% at 5,220 employees) — large workforce periods coincide with economic expansion when term deposits are less attractive relative to other investment options.
- **Consumer confidence index (avg −40.50)** shows a positive correlation with conversion — the less negative CCI (closer to 0), the higher the conversion rate. Higher confidence means customers are more willing to commit to a product.

- **Medium rate environment (287 contacts, 44.25% conversion)** is a tiny but extraordinarily productive window — each contact in this environment yields nearly 4× the return of a low-rate contact. These windows should trigger an immediate campaign surge.
- **The macro indicators are mutually correlated** — EVR, number of employees, CPI, and interest rate all move together, describing a single underlying economic regime. Any one indicator can serve as a reliable proxy for campaign timing decisions.
- **Avg interest rate of 3.62%** across the campaign window contextualises the dataset as spanning both pre- and post-crisis recovery periods, making its findings more generalisable to future rate-cycle planning.

Part IV

Decisions

Translating evidence into action — what to do, what's at risk, and what comes next.

010 Key Insights
011 Business Recommendations
012 Limitations & Next Steps
013 Contribution Matrix

IV

SECTION 10

Key Insights

Twelve decision-oriented findings, each tied back to a specific statistical result or KPI. Use these as the canonical headline list when summarising for stakeholders.

01

Baseline conversion is 11.27% — every recommendation in this report is benchmarked against this rate.

02

Previous-campaign success is the single strongest predictor. Repeat-success customers convert at 65.11% vs 8.83% for cold contacts (Chi-Square = 4,113.95, 6.9× lift).

03

Conversion is U-shaped in age. Senior 55+ converts at 46.9% and Young ≤25 at 21.0%, while the 36–45 cohort lags at 8.4% — the largest segment by volume is also the worst by yield.

04

Students and retired customers dominate by occupation; blue-collar workers underperform.

05

Cellular outperforms landline ~3× (14.74% vs 5.23%, Chi-Square = 862.08).

06

Overcalling destroys yield. Conversion drops monotonically from 13.04% (1 call) to 5.49% (6+ calls). Logistic-regression coefficient on *num_contacts_campaign* is negative.

07

March, September and December are the highest-converting months (50.55%, 44.91%, 48.90%) — but May is the highest-volume month, indicating a clear volume / yield mismatch.

08

Macro-economic regime is decisive. Low-rate windows convert at 24.02% vs 4.83% in high-rate windows (F = 1,977.46).

09

Call duration is the strongest behavioural signal of intent (r = +0.41; logistic OR = 3.23) — qualified longer calls correlate with successful conversion.

10

Recently contacted customers respond better than cold leads — recency bucket matters more than the raw days-since-last-contact number.

11

Macro indicators move together. Employment-variation rate, CPI, interest rate and employee count are highly inter-correlated.

12

The logistic-regression baseline reaches 90.8% accuracy using just six features, indicating that a lightweight propensity model is feasible with no further data.

SECTION 11

Business Recommendations

Five ranked business recommendations, each anchored to a specific statistical result. Each card opens with the data-driven question that the recommendation directly answers. Priority reflects expected business impact relative to implementation cost.

01 Re-rank the call list by prior-success and demographic flags

If prior-success customers convert at 65.11%, why aren't they the first call every morning?

RATIONALE & EVIDENCE

Move prior-success customers and Senior 55+ / Young ≤25 cohorts to the top of every daily call list. **Evidence:** 6.9× lift on prior success (Chi-Square = 4,114); 4.2× lift on Senior 55+ vs Mid 36–45.

EXPECTED BUSINESS IMPACT

Lift expected conversion in the re-contact tier from ~11% toward 30–50% on those targeted segments; reduces wasted calls in the 36–45 majority.

02 Cap campaign call-intensity at 3 per customer

If conversion drops to 5.49% after 6+ calls, why are agents still dialling the same number repeatedly?

RATIONALE & EVIDENCE

Each marginal call after the third reduces yield (13.04% → 5.49% by call 6+). Build a soft-stop into the dialler.

EXPECTED BUSINESS IMPACT

Reduces agent hours on low-yield repeat calls; improves customer experience and compliance posture; redirects effort to high-propensity prospects.

03 Shift channel mix toward cellular

If cellular converts at nearly 3× landline, what is the cost of still routing calls through telephone?

RATIONALE & EVIDENCE

Cellular converts ~3× landline (14.74% vs 5.23%, Chi-Square = 862). Phase down landline outreach where cellular is available.

EXPECTED BUSINESS IMPACT

Direct uplift on conversion rate of channel-mixed campaigns; potential cost savings if cellular dialling is cheaper per minute.

04 Time campaigns to low-rate windows and to March / September / December

If May is the busiest contact month yet converts at only 6.44%, who decided to run the campaign then?

RATIONALE & EVIDENCE

Run heavy campaigns when interest_rate is in the Low band (24.02% conversion) and in the three highest-yield months identified by EDA.

EXPECTED BUSINESS IMPACT

Macro timing alone is associated with a ~5× swing in conversion rate; even partial alignment to low-rate windows is highly leveraged.

05 Stand up a lightweight propensity score using the six logistic-regression features

If a six-feature model already achieves 90.8% accuracy, why is the campaign still using an undifferentiated contact list?

RATIONALE & EVIDENCE

Productionise the existing model (call_duration is post-hoc; use the other five as a pre-call score) and feed it into the dialler.

EXPECTED BUSINESS IMPACT

Baseline model already at 90.8% accuracy — gives the campaign manager a reproducible, explainable scoring layer with no new data dependencies.

SECTION 12

Limitations & Next Steps

Data Limitations

- **Public benchmark dataset, not a live feed.** Findings reflect one Portuguese bank's historical campaigns and may not generalise to other geographies, products or eras.
- **No native region field.** Our NUTS II *region* column is probability-imputed from *job_type*, so regional dashboards should be read as illustrative rather than ground-truth geography.
- **'Unknown' as a category** in several demographic fields limits the precision of demographic segmentation.
- **Severe class imbalance** (11.27% positives) — accuracy is a misleading metric on its own; precision/recall/F1 would be needed for a deployable model.
- **Capping num_contacts_campaign at 10** trades a small amount of tail signal for stable aggregations.

Method Limitations

- Logistic regression is intentionally simple — assumes additive log-odds and no feature interactions. Tree-based models would likely outperform.
- **call_duration_sec is post-hoc** — it cannot be used in a pre-call propensity score, even though it is the strongest correlate.
- Macro features are highly inter-correlated, which inflates standard errors and makes individual coefficients harder to interpret in isolation.
- Hypothesis tests confirm *association*, not *causation* — uplift would need to be established with controlled A/B testing.

Suggested Future Work

- **Replace the logistic baseline with a gradient-boosted model** (XGBoost / LightGBM) and report precision-recall metrics with class-weighted loss.
- **Add channel cost** per contact to compute a true cost-per-acquisition KPI rather than raw conversion rate.
- **Run a small A/B test** on the call-cap-at-3 recommendation before rolling out — this is the lowest-risk, highest-confidence change.
- **Incorporate customer-level transaction data** (where available) to add wealth and engagement signals beyond demographics.
- **Build a real region field** from genuine address data; deprecate the imputed NUTS II column.
- **Deploy the propensity score back to the dialler** as a daily-refreshed scoring layer.

SECTION 13

Contribution Matrix

This table must match evidence in GitHub Insights, PR history, and committed files.

Team Member	Dataset and Sourcing	ETL and Cleaning	EDA and Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT
Abhayanth K	Support	Support	Support	Support	Support	**Owner**	Support
Kasula Lalithendra	Support	**Owner**	**Owner**	Support	Support	Support	Support
Jagruthi Pulumati	**Owner**	**Owner**	Support	Support	**Owner**	Support	Support
Sachin Jaiswal	Support	**Owner**	Support	Support	Support	Support	**Owner**
Ankit Raj Singh	Support	Support	Support	**Owner**	Support	Support	Support
Varun Sharma	—	—	—	—	—	Support	—

Declaration: *We confirm that the above contribution details are accurate and verifiable through GitHub Insights, PR history, and submitted artifacts.*

Team Lead Name: Jagruthi Pulumati

Date: 29 April 2026